AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2012

Chemistry Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL EIGHT questions. Questions 6, 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 4 Marks)

a. Environmental chemists have expressed their concerns about the depletion of ozone layer. Increased quantities of chlorofluorocarbons are carried by convection of currents to the stratosphere, where they absorb ultraviolet radiations and react with ozone.

$$CFCl_3 \rightarrow CFCl_2^{\bullet} + Cl^{\bullet}$$
$$Cl^{\bullet} + O_3 \rightarrow ClO^{\bullet} + O_2$$
$$ClO^{\bullet} + O \rightarrow Cl^{\bullet} + O_2$$

Classify $Cl^{\bullet}$ and O_2 species as atom, molecule, free radical or ion. Give a reason for the identification of $Cl^{\bullet}$. (3 Marks)

Identification of $Cl^{\bullet}$: _____

Identification of O_2: _____

Reason for the identification of $Cl^{\bullet}$: _____

_____

_____

b. Cl_{17}^{35} and Cl_{17}^{37} are two isotopes of chlorine. What do you understand by the term isotopes? (1 Mark)

_____

_____

Q.2. (Total 5 Marks)

a. A local scientist had discovered an imaginary element having atomic number 18. She wanted to place it in the modern periodic table for which she wanted to know the group and period number of the imaginary element. (4 Marks)

i. Identify the period and group number of the imaginary element with the help of electronic configuration.

Electronic Configuration _____

Period Number _____

Group Number _____

ii. Write ONE characteristic of the above identified group.

_____

_____

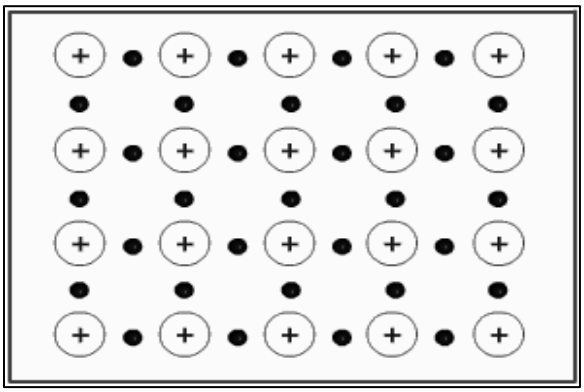
b. State in which order scientists have arranged the elements in a period in the modern periodic table. (1 Mark)

_____

_____

Q.3. (Total 6 Marks)

a. Copper metal is used to make electrical wires to conduct electricity. A general structure of the metal is shown below.



Answer the following questions with reference to the given figure.

i. Why are copper metal wires used to conduct electricity? (1 Mark)

_____

_____

ii. Which type of force of attraction is present in the given structure? Also give ONE characteristic of it. (2 Marks)

_____

_____

_____

_____

b. Explain the relationship of boiling point and vapour pressure. Also state the effect of external pressure on boiling point. (3 Marks)

_____

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Q.4. (Total 6 Marks)

a. Silver plating is done on copper utensils to make it shiny. The following redox reaction occurs during silver plating.

$$2AgNO_{3(aq)} + Cu_{(s)} \rightarrow Cu(NO_3)_{2(aq)} + 2Ag_{(s)}$$

i. Identify the reduced species in the given reaction. (1 Mark)

_____

_____

ii. Identify the change in the oxidation state of the reduced species. (1 Mark)

_____

_____

iii. Identify the reducing agent in the given reaction. Give a reason for your identification. (2 Marks)

_____

_____

_____

_____

b. A doctor prescribed milk of magnesia, a suspension of magnesium hydroxide, to a patient in order to be used as a neutralizer for stomach acidity.

How would you determine that milk of magnesia is a suspension? Give TWO reasons. (2 Marks)

_____

_____

_____

_____

Q.5. (Total 4 Marks)

Justify the following statement.

'The first ionization energy of group IIA is higher than group IA whereas the second ionization energy of group II A is lower than group I A.'

Q.6. (Total 5 Marks)

EITHER

a. A compound **X** contains 72 % carbon, 4 % hydrogen and 24 % oxygen. The molecular weight of the compound is 130 g. Calculate its empirical and molecular formula.

OR

b. Draw the atomic structure of the following elements and their ions. Mention the number of subatomic particles in each case. Also specify which of the following elements form a cation or an anion. (Hint: Fluorine ($^{19}_{9}F$) and Aluminium ($^{27}_{13}Al$))

i. Fluorine atom and fluoride ion ii. Aluminium and aluminium ion

Q.7. (Total 5 Marks)

EITHER

a. Describe the formation of a covalent bond between two unlike non- metallic elements. Also give the diagrammatic representation of the formation of the bond.

OR

b. Define allotrope and describe any TWO crystalline allotropic forms of carbon with reference to their crystalline structure.

Q.8. (Total 5 Marks)

EITHER

a. Define solubility and discuss any TWO factors which affect solubility with the help of an example.

OR

b. Describe the extraction of aluminium from alumina by using the process of electrolysis.

END OF PAPER

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